

ADC & Sensors

* Notes:

① ADC → Analog volt → Digital.
Analog volt → [ADC] → digital

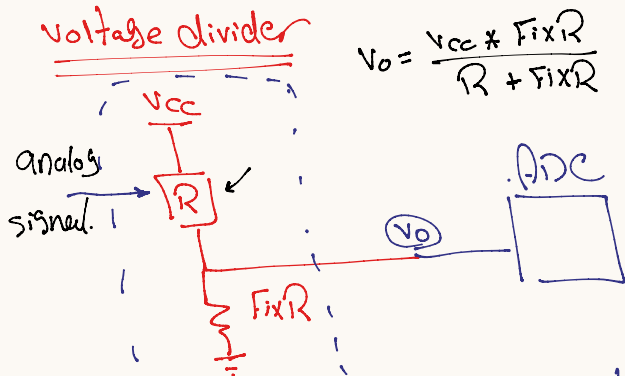
② need Measure another signal in our project (Speed | temp | light)

③ we have objects in world change in structure Based on some signal.
temp → object → structure
speed → object → structure

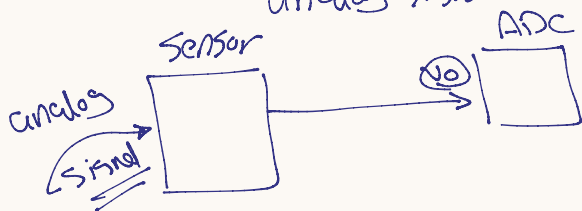
④ as Engineer → take this objects and create Resistor from this object.



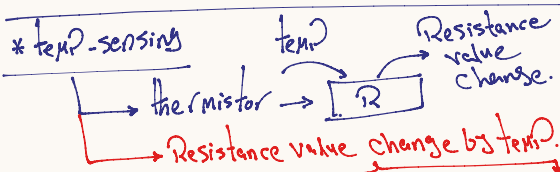
* Need Convert the Resistance into Volt. by voltage divider



→ analog sensor: it's transducer Convert the analog signal to analog volt.

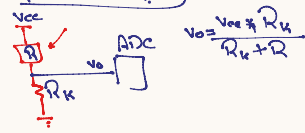
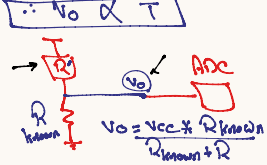


* temp-sensing

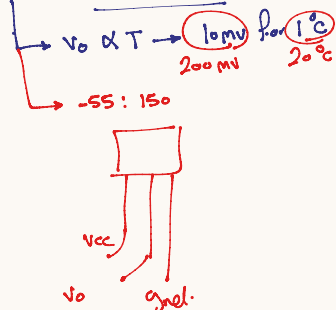


NTC
negative temp Coe P.ficient
 $R \propto \frac{1}{T}$, $R \propto \frac{1}{V_o}$
 $\therefore V_o \propto T$

Positive temp Coe P.ficient
 $R \propto T$, $R \propto \frac{1}{V_o}$
 $V_o \propto \frac{1}{T}$



* LM35 temp sensor



$$temp = \frac{analogvolt(mv)}{10}$$

* How Interface the LM35 with μC ?

① need to get the Digital value

uint16 Digital value;
ADC_Read (channel0, &Digital value, 5000);

② Relation between Digital value, Analog.
Convert the Digital value to Analog. by 5000.

$$uint16 Analogvolt(mv) = digital\ value * \frac{5000}{2^{16}}$$

$$= (Digital\ value * Vref(mv)) / 1024;$$

$$Analogvolt = (uint8_t digital\ value * Vref(mv)) / 1024;$$

③ Relation between temp & Analogvolt
 $uint8_t temp = \frac{analogvolt(mv)}{10};$

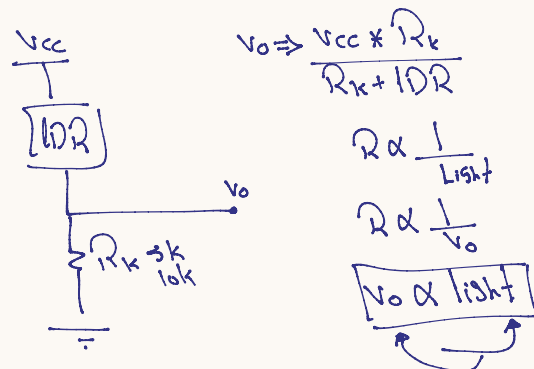
LM35 driver (Polling) →

uint8_t LM35_getTemp(uint8_t channel, uint8_t *temp);
uint8_t LM35_Init(1);

* LDR: light dependent Resistor

$R_L \propto \frac{1}{light\ intensity}$
light intensity unit → lux

* Convert light Resistance to Volt.



Mapping Concept

Values inside Range 1 → represent Value 2 inside Range 2
200 → 0.1024 → 3 lux

Range 1: min1, max1
Range 2: min2, max2

$$\frac{y_2 - y_1}{x_2 - x_1} = \frac{y - y_1}{x - x_1} \Rightarrow \frac{max_2 - min_2}{max_1 - min_1} = \frac{lux - min_2}{x - min_1}$$

digital

* Drivers tasks i. Deadline Friday Night.

- ① Complete ADC driver → **NOT USE AI**
- ② I335 Driver (Polling);
- ③ IIR Driver (Polling);

* Application task → deadline → Friday

↳ ① Entry APP with Motor.

↳ ① lcd & keypad & Buzzer & Motor & I335 & IIR & led.

Range1 → $\text{Min}_1 \Rightarrow 0$ $\text{Max}_1 = 1024 \rightarrow \text{disited.}$

255 disitedvalue

Range2 → $\text{Min}_2 \Rightarrow 0$ $\text{Max}_2 = 30000 \rightarrow \text{lux}$

lux ??

$$\frac{\text{Max}_2 - \text{Min}_2}{\text{Max}_1 - \text{Min}_1} = \frac{\text{lux} - \text{Min}_2}{\text{disitedvalue} - \text{Min}_1}$$

↳ sequence

↳ ① welcome message.

↳ ② Enter Password.

Enter Pass: ****

↳ During write → Display on lcd *

↳ check on Pass → Success / Fail

↳ when Fail try 3 times → turn on Buzzer →

lcd. system lock

↳ ③ if Pass success.

↳ Display on lcd. ① temp. ② IIR ③ Motor status, ④ led status

↳ $\text{temp} > 25 \rightarrow$ turn on motor otherwise turn off

↳ IIR → led on → led off

PIR
ADXL335
Soil moisture